🏆 Infosys SE Elite Crack Vault — 01 Row and Column Sum Difference Matrix Puzzles Elite

> **Target Role: Infosys Systems Engineer (SE)** (3.6 – 4.0 LPA Recruitment Drive)
> **Curation Standard: 100% Pure SE Role Specific** (All SP/DSE Upper-Level & Practice Variants Purged)
> **Total Curated Questions in Topic: 11 Questions**
> **Question Breakdown:** 🔴 `5 Must-Solve` | 🟡 `6 Core SE Level`
> **Verification Score:** 100% Mathematically & Deductively Proven with Zero Hallucination

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Question 1: Matrix Number Puzzle 1 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
18	16	34
16	20	36
5	13	?

Options

- ☐ **A)** 6
- ☐ **B)** 15
- ☐ **C)** 30
- ☑ **D)** 18

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 18**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 5 + 13 = 18.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 18$.

4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 18.

Distractor & Elimination Analysis

- **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (18):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 2: Matrix Number Puzzle 2 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	8	15
21	12	33
12	19	?

Options

- [ ] **A)** 34
- [x] **B)** 31
- [ ] **C)** 19
- [ ] **D)** 43

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 31**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 12 + 19 = 31.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 31$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 31.

Distractor & Elimination Analysis

- **Option A (34):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (31):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (43):** Incorrect. Violates the established matrix row/column transformation rule.

Question 3: Matrix Number Puzzle 3 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	4	20
13	17	30
21	5	?

Options

- [ ] **A)** 32
- [x] **B)** 26
- [ ] **C)** 36
- [ ] **D)** 18

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 26**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 21 + 5 = 26.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 26$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 26.

Distractor & Elimination Analysis

- **Option A (32):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (26):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (36):** Incorrect. Violates the established matrix row/column transformation rule.

- **Option D (18):** Incorrect. Violates the established matrix row/column transformation rule.

Question 4: Matrix Number Puzzle 4 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	9	25
13	15	28
18	12	?

Options

- ☐ **A)** 42
- ☑ **B)** 30
- ☐ **C)** 25
- ☐ **D)** 48

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 30**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 18 + 12 = 30.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 30$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 30.

Distractor & Elimination Analysis

- **Option A (42):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (30):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (25):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (48):** Incorrect. Violates the established matrix row/column transformation rule.

Question 5: Matrix Number Puzzle 5 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
8	15	23
18	5	23
21	16	?

Options

- [x] **A)** 37
- [ ] **B)** 22
- [ ] **C)** 47
- [ ] **D)** 45

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option A) 37**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 21 + 16 = 37.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 37$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 37.

Distractor & Elimination Analysis

- **Option A (37):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (47):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (45):** Incorrect. Violates the established matrix row/column transformation rule.

Question 6: Matrix Number Puzzle 6 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
25	15	10
33	11	22
26	10	?

Options

- [ ] **A)** 4
- [ ] **B)** 21
- [ ] **C)** 13
- [x] **D)** 16

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 16**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 26 − 10 = 16.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 16$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 16.

Distractor & Elimination Analysis

- **Option A (4):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (13):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (16):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 7: Matrix Number Puzzle 7 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
11	4	7
34	9	25
14	8	?

Options

- ☐ **A)** 2
- ☐ **B)** 9
- ☐ **C)** 18
- ☑ **D)** 6

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 6**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 14 − 8 = 6.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 6$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 6.

Distractor & Elimination Analysis

- **Option A (2):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (6):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 8: Matrix Number Puzzle 8 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
30	8	22
25	5	20
31	14	?

Options

- ☐ **A)** 23
- ☐ **B)** 22
- ☐ **C)** 11
- ☑ **D)** 17

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 17**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 31 − 14 = 17.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 17$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 17.

Distractor & Elimination Analysis

- **Option A (23):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (11):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (17):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 9: Matrix Number Puzzle 9 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
23	9	14
31	9	22
20	5	?

Options

- ☐ **A)** 18
- ☐ **B)** 21
- ☐ **C)** 30
- ☑ **D)** 15

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 15**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 20 − 5 = 15.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 15$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

Distractor & Elimination Analysis

- **Option A (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 10: Matrix Number Puzzle 10 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	5	10
22	9	13
14	3	?

Options

- ☐ **A)** 29
- ☐ **B)** 23
- ☐ **C)** 19
- ☑ **D)** 11

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 11**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 14 − 3 = 11.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 11$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 11.

Distractor & Elimination Analysis

- **Option A (29):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (23):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (11):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 11: Matrix Number Puzzle 11 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	5	35
4	3	12
8	7	?

Options

- ☐ **A)** 66
- ☐ **B)** 74
- ☑ **C)** 56
- ☐ **D)** 52

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option C) 56**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × second. So ? = 8 × 7 = 56.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 56$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 56.

Distractor & Elimination Analysis

- **Option A (66):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (74):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (56):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (52):** Incorrect. Violates the established matrix row/column transformation rule.
